CSE 561 Midterm Exam

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I. Question

Define the relations **gorumce(X, Y)**, **elti(X, Y)**, **bacanak(X, Y)**, **kayin(X, Y)** in terms of parent, sister, married, female, male relations.

II. Question

Define relation **max3(X, Y, Z, M)** where M is the maximum of X, Y, Z.

III. Question

Define relation **fib(X, Y)** where Y is the X.th fibonacci number.

IV. Question

Define relation **fahr(X, Y)** where X and Y are lists and Y contains the Fahrenheit converted values of celcius values in X.

V. Question

Write a program which finds the number with 6 digits having the property: When multiplied 2, 3, 4, 5, and 6 the digits of the number remain the same, only the order of the digits changes.

VI. Question

Write a predicate **palsublist(L, Sublist)** which finds all palindrome sublists of a list L.

VII. Question

We have the following information: Emrah loves Selin, Selin loves Emrah, Nurgl and Onat, Nurgl loves Cemal, Onat and Sibel, Cemal loves Emrah and Selin, Onat loves Selin and Nurgl. Two people who love each other can go to cinema together. Two people who can go to cinema together and are of opposite sex can marry. Write a prolog program that finds the pairs who can marry.

VIII. Question

We want to schedule courses in a small school according to the following criteria:

- There are four courses: algebra, geometry, calculus, and analysis.
- There are four teachers: Nurgul, Rifki, Emrah, and Selin.
- All courses must be taught.
- Each teacher must teach a different course.
- There is a single room and all lectures must be assigned to this room.
- The school is open only on Saturdays. Each course is taught just 1 hour per week. The times are from 1 to 8 (i.e. between 9:00 and 17:00).
- Nurgul can teach algebra and geometry; Rfk can teach calculus and analysis; Selin can teach algebra, calculus, and analysis; Emrah can teach any course.
- Rifki wants to teach only on 1st, 2nd, or 3rd hours; Selin wants to teach only between 5th and 8th hours (i.e. in the afternoon).
- If Nurgul teaches algebra, she wants to teach it between 1st and 4th hours; if she teaches geometry, she wants to teach it between 5th and 8th hours.

Write a Prolog program that, given a candidate schedule as input, determines whether the schedule is valid or not. The syntax of the input is [[c1,t1,h1],[c2,t2,h2],[c3,t3,h3], [c4,t4,h4]]. We assume that the input is error-free (i.e. it is not necessary to check the contents of the input list): There are exactly four sublists, each course appears in exactly one sublist, each teacher appears in exactly one sublist, all hours are between 1 and 8 and there is no duplication of hours. (That is, you need to check only the last three constraints g, h, and i.) For instance, if the input is [[nurgul,algebra,1], [selin,analysis,5], [emrah,geometry,3], [rfk,calculus,2]], the program will decide that this is a valid schedule. If the input is [[nurgul,calculus,1], [selin,analysis,5], [emrah,geometry,3], [rifki,algebra,2]], it will decide that this is an invalid schedule.

IX. Question

Write a Prolog program that returns the middle element of a given list. The program will take a single list parameter.

X. Question

Write a Prolog program that, given an integer N ¿ 2, generates all the prime numbers between 3 and N. For instance, if the program is called as prime(20) it will output 3,5,7,11,13,17,19 (or, in reverse order).